Cells and Gauge

File: gauge.kicad_sch

Sleep Current and 3V3

File: power.kicad_sch

MCU and NTCs

File: mcu.kicad_sch

USB Port 1

IRQ
EN_5V
ADCIN1
ADCIN2
LDO_3V3

IRQ1
EN_5V1
P1_ADCIN1
P1_ADCIN2
P1_LDO_3V3

File: usb_port.kicad_sch

USB Port 2

IRQ
EN_5V
ADCIN1
ADCIN2
LDO_3V3

IRQ2
EN_5V2
P2_ADCIN1
P2_ADCIN2

File: usb_port.kicad_sch

P1_LDO_3V3
P1_ADCIN2
P2_ADCIN1
P2_ADCIN2
P1_ADCIN1
R1 0R
R2 0R
R3 0R
R4 0R
GND
GND
GND
GND

TPS25751 ADDRESS / DEAD-BATTERY STRAPS (SafeMode)
Port 1: R1 ADCIN1 = LDO_3V3, R2 ADCIN2 = GND -> I2C 0x20
Port 2: R3 ADCIN1 = GND,   R4 ADCIN2 = GND -> I2C 0x21
Gauge BQ40Z50-R2 = 0x0B on the same SDA/SCL bus (STM32WB I2C1, 100 kHz)

MONOLITH DEV BOARD - SHEET INDEX
2 Cells and Gauge   4S holders / pack input, BQ40Z50-R2, protection FETs, SMBus
3 Sleep Current and 3V3 R_MEAS + 2x INA190 to test loops, HT7550-1 amp supply, TPS62177 3V3
4 MCU and NTCs   STM32WB1MMC, debug headers, board NTCs TH1-TH4, LED
5 USB Port 1 / 6 USB Port 2   (same sheet file) TPS25751D + BQ25756 + TPS62933 per port
Rails (power symbols): PACK_P, 3V3, GND, CELL_N (gauge VSS only)
Spec: docs/devboard/circuit-description.md   Changes/open items: DevBoard/README.md

REVISION LOG
A0 2026-10-11 First draft from circuit description draft 5. Author: Claude Code. Reviewer: none yet.
A1 2026-10-11 Layout cleanup: blocks, power symbols, notes, title blocks. Author: Claude Code. Reviewer: none yet.
A2 2026-10-11 Readability pass: no overlapping text, sheet-pin names inside boxes. Author: Claude Code. Reviewer: none yet.
A3 2026-10-11 Wiring pass: wires instead of labels inside each block. Author: Claude Code. Reviewer: none yet.
A4 2026-10-12 Polymer bulk caps, PWR_FLAGs at rail sources, design-review fixes
                (docs/devboard/schematic-review-2026-10-12.md). Author: Claude Code. Reviewer: 3-agent review.
A5 2026-10-13 Review 2 decisions: CHG_INH fails safe (R417/R517 47k pull-up to 3V3, MCU drives low to charge),
                MCU IWDG in hardware mode; BAT fed before F101 (TI fig. 9-12); C421/C521 100 uF; LRB520S, BSS138,
                C109 4.7n, PPHV/VBUS caps, GND/probe test points, JLC numbers on all R/C (review 2);
                J201/J202 SMAs replaced by OUT_LO/OUT_HI loops TP201/TP202 + GND loops TP208/TP209. Author: Claude Code.
A6 2026-10-14 Branch hand-review: second-level protection U102 BQ294700 + F102 WSFB1514 (TI fig. 9-11), F101 deleted;
                D103 fitted; U202 INA190A1 JLC-placed; R405-R408 removed; USB-C liquid detection (U405/U505 TPD2S300,
                Q403-Q406, R440/R540 DCP short, D403/D503 removed). Review 3: TPD2S300 FLT -> GPIO3 (R441), liquid-detect
                gate pulls R442/R443, BQ25756 gate/BTST OR R444-R449 + DNP snubbers, U401 symbol GPIO11 bidirectional.
                Cost-down (docs/devboard/cost-down-2026-10-14.md): L401/L501 SMMS1040, L402/L502 APS0530, L201 FNR4030S,
                U203 HT7550-1, 22u 1206 battery bank, basic/preferred passives, Q103 AO3400A, D301 red, R431 solder
                jumper, 34 test loops -> pads, headers hand-soldered, U301 consigned.
                Charger half-bridges IAUC45N04 -> 4 discrete PGQ04N070 per port (Q401/Q402 + new Q407/Q408).
                6-layer 1.6 mm board (0.3 mm vias): U405/U505 TPD2S300 -> TPD4S201 (CC + SBU), D402/D502 removed,
                J401/J501 -> TYPE-C 16PCB1.6 IP68.
                Author: Logan (hand review) + Claude Code.

Not reviewed, laid out, built or tested
Spec + open items: DevBoard/README.md
Sheet index, PD address straps
Monolith 100 W battery bank — dev board

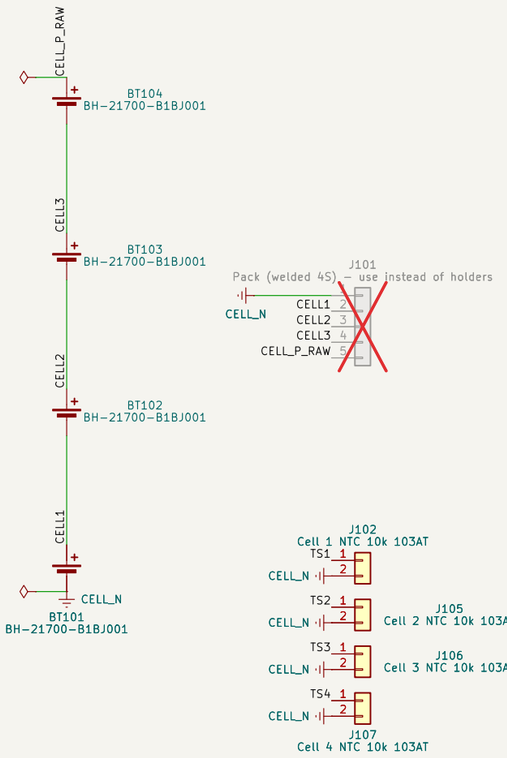
Sheet: /
File: MonolithDev.kicad_sch

Title: Monolith dev board — top level

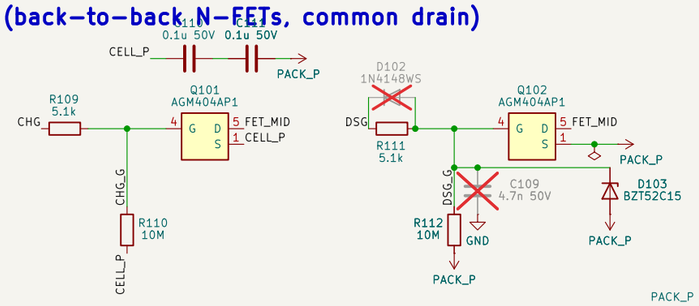
Size: A3	Date: 2026-10-11	**Rev: A (draft)**
KiCad E.D.A. 10.0.7		Id: 1/6

CELL STACK (4S)

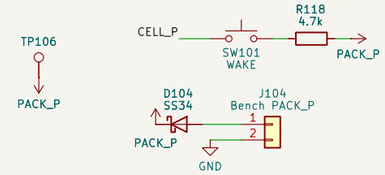
Holders BT101–BT104 OR welded pack on J101 — never both.
J101 is DNP. Check each cell at TP102–TP105 and the stack at TP108 before waking the gauge (Q101/Q102 stay off until FET_EN is set, so nothing loads the cells until then).

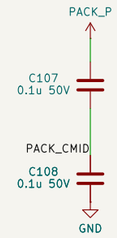


PROTECTION FETs (back-to-back N-FETs, common drain)

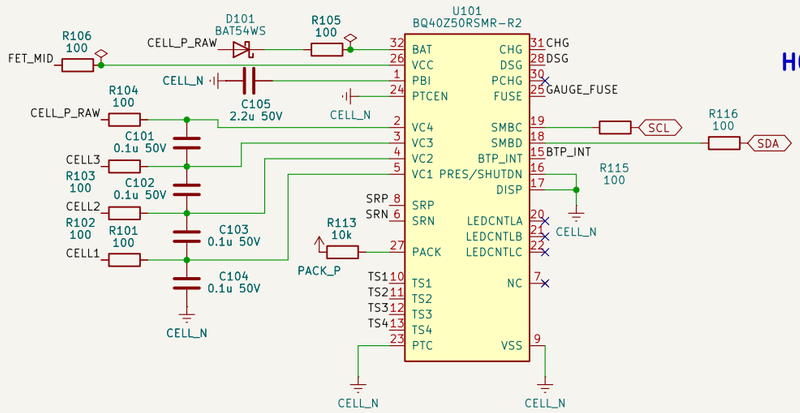


WAKE + BENCH SUPPLY

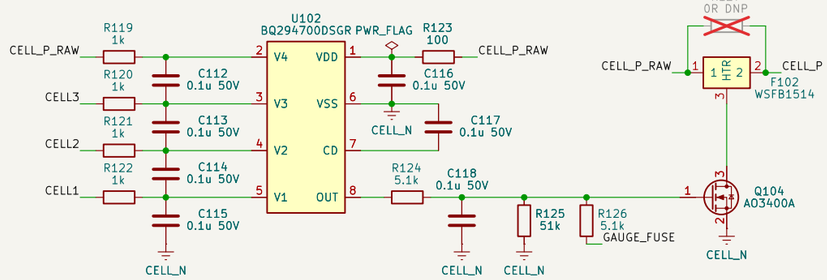




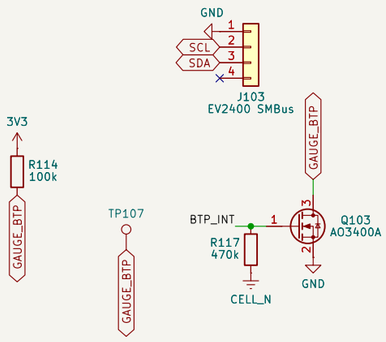
GAS GAUGE BQ40Z50-R2 (TI fig. 9-1)



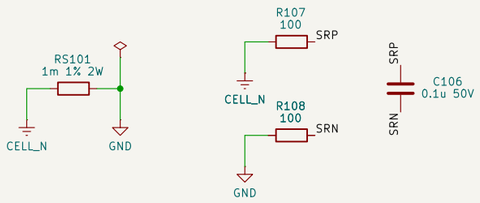
SECOND-LEVEL PROTECTION BQ294700 + SCP (TI fig. 9-11)



HOST SMBus (gauge 0x0B) / EV2400



LOW-SIDE CURRENT SENSE



TEST POINTS / POWER FLAGS



NOTES
1. CELL_N = gauge VSS (cell 1N tab). GND = system ground, on the far side of RS101.
2. Kelvin-route SRP/SRN from the RS101 pads; R107/R108/C106 at the IC.
3. VC taps are separate thin traces from the cell end of each holder contact;
   VC4 from CELL_P_RAW (before F102) so fuse drop is not read as cell voltage.
4. TS1–TS4: one wired cell NTC each (J102, J105–J107, one per cell); insulated sensors only (cans 0–12.6 V above VSS, TS 2.1 V max). PTC, PTCEN, PRES, DISP tied to VSS per datasheet.
5. VCC fed from FET common drain via R106 (TI R13). PACK sense via R113 10k (TI R15).
6. BAT fed from CELL_P_RAW (cell top, before F102) via D101 + R105 (TI fig. 9-12).
7. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
8. Q103 buffers BTP_INT (its high level has no maximum: PA6 takes 4 V unpowered): GAUGE_BTP is active low.
9. C109 + D102 (DNP): fit if the DSG turn-on inrush into PACK_P trips SCD. D103 (fitted) clamps Q102 VGS at 15 V in a hard short.
10. Tape the J102 NTC to BT101: a 21700 can is its cell's negative terminal and TS1 is rated 2.1 V.
11. SW101 wakes the gauge from SHUTDOWN without USB; J104 feeds PACK_P from a current-limited bench supply.
12. J103 matches the EV2400 SMBus port: 1 VSS, 2 SCL, 3 SDA, 4 VOUT (not used). Fit cells before plugging in USB.
13. ESD caps (TI): C110+C111 in series across the FET pair (fig. 9-2 C1/C2), C107+C108 across PACK+/PACK- (sec. 9.2.2.1.5).
14. Second level (TI fig. 9-11): U102 BQ294700 (4.35 V/cell, CD 0.1u = 1-2 s) or the gauge FUSE pin turns Q104 on;
    F102's heater then blows its fuse, permanently opening the cell top. F102 is also the pack's only overcurrent fuse (no F101).
    R127 (DNP) is a bench bypass for a blown F102 only: never fitted on a pack in use.
15. Never insert or remove a cell with USB or J104 power on: an empty holder slot sees PACK_P minus the other cells
    (> 4.35 V), U102 trips after 1-2 s and F102 blows. Gauge data flash: COV Delay 1 s, Min Blow Fuse Voltage 10500 mV.

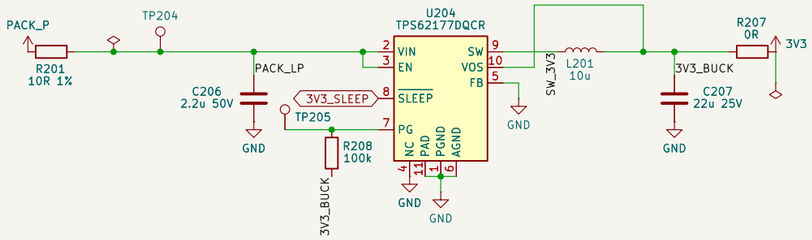
Not reviewed, laid out, built or tested
Spec + open items: DevBoard/README.md
4S input, BQ40Z50-R2 gauge/protector, protection FETs
Monolith 100 W battery bank — dev board

Sheet: /Cells and Gauge/
File: gauge.kicad_sch

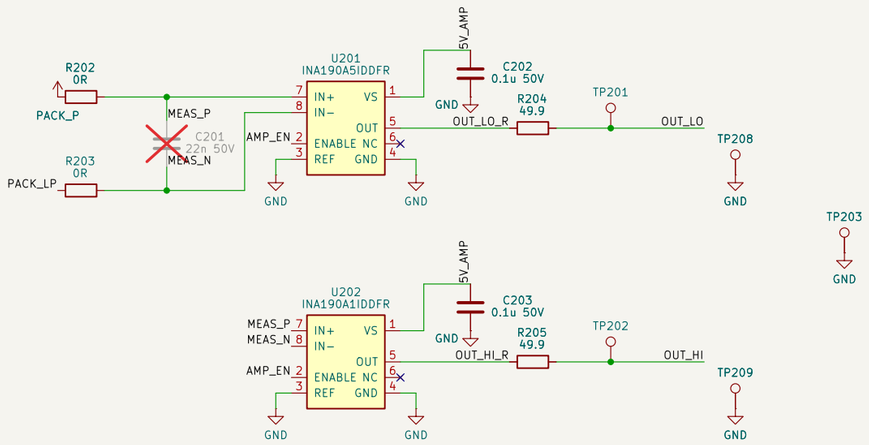
Title: Cells and gauge

Size: A3	Date: 2026-10-11	**Rev: A (draft)**
KiCad E.D.A. 10.0.7		Id: 2/6

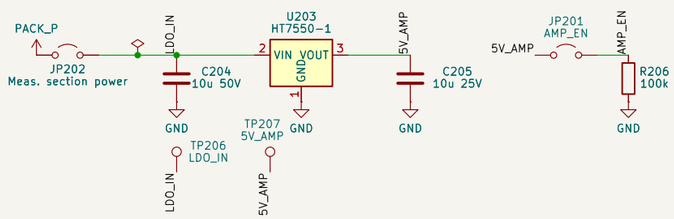
3.3 V ALWAYS-ON RAIL (TPS62177, 1 MHz, sleep mode under MCU control)



SLEEP-CURRENT MEASUREMENT (both amps across R201 = R_MEAS)



AMPLIFIER SUPPLY (5 V LDO, fed from PACK_P so amp current bypasses R_MEAS)

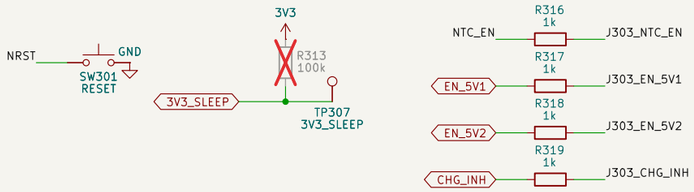


NOTES
1. Low-current branch: PACK_P -> R201 10R -> PACK_LP -> TPS62177 -> 3V3 (MCU + both TPS25751). Gauge BAT is NOT on this branch (TI: gauge must stay powered with FETs off).
2. Kelvin-connect R202/R203 to the R201 pads.
3. U201 gain 500: 5 mV per uA, -0.99 mA full scale. U202 gain 25: 0.25 V per mA, -20 mA full scale.
4. R202/R203/C201 fitted 0R/0R/DNP to start; 100R/100R/22nF gives a 36 kHz input filter.
5. Probe OUT_LO/OUT_HI at TP201/TP202, ground lead on TP208/TP209 beside them (1 MOhm scope input). R204/R205 49.9R isolate the INA190 outputs from probe/cable capacitance (1 nF max).
6. SLEEP low = sleep mode (internal pull-down); MCU PB4 drives it high for full load.
7. R207 breaks the 3V3 rail for debugging.
8. U201/U202 are hand-soldered (DigiKey/Mouser); JLC has no stock. Not in the JLC BOM/CPL.
9. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
10. U203 EN is left open (internal 300 nA pull-up enables it). EN must stay below 6.5 V: never tie it to LDO_IN.
11. U204 pin 4 (NC) goes to GND, as TI recommends.

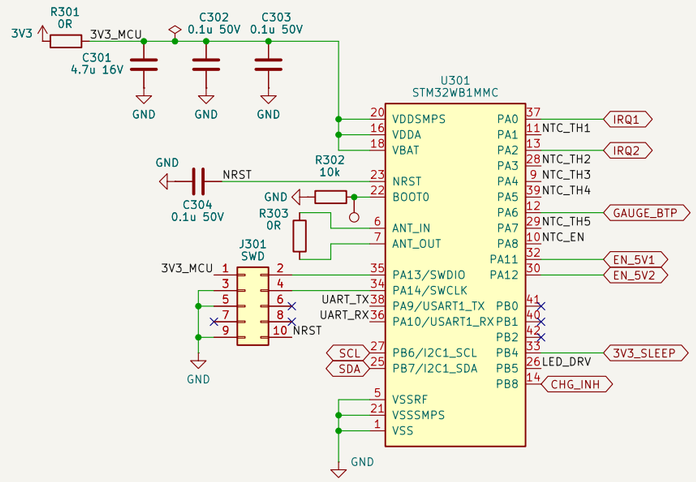
Not reviewed, laid out, built or tested
Spec + open items: DevBoard/README.md
R_MEAS + INA190 x2, HT7550-1, TPS62177
Monolith 100 W battery bank — dev board

Sheet: /Sleep Current and 3V3/		
File: power.kicad_sch		
Title: Sleep-current measurement and 3.3 V rail		
Size: A3	Date: 2026-10-11	**Rev: A (draft)**
KiCad E.D.A. 10.0.7		Id: 3/6

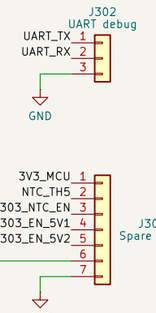
BRING-UP AIDS



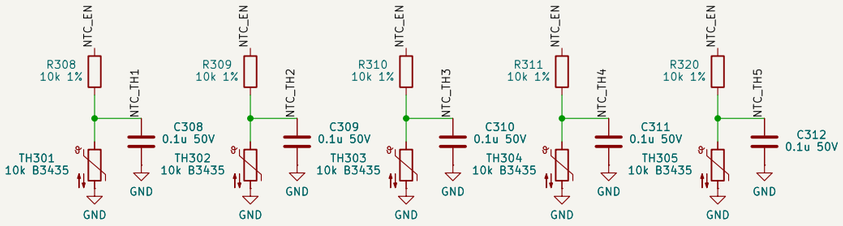
BLUETOOTH MCU STM32WB1MMC (per ST DS14096 fig. 5)



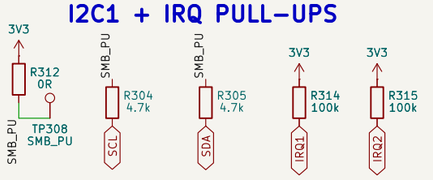
DEBUG HEADERS



BOARD NTCs (TH1–TH4, divider powered only while sampling)

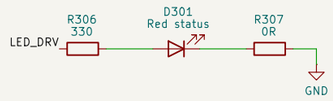


I2C1 + IRQ PULL-UPS



NOTES
1. U301 at a board edge, as on ST's reference PCB (DS14096 fig. 6): copper-free notch past the antenna end.
2. C301–C303 at the VDD pads; C304 at NRST.
3. PB0/PB1/PB2 left unconnected (ST: RF-sensitive pins). NTC_EN = PA8, EN_5V1 = PA11, EN_5V2 = PA12.
4. R303 0R joins ANT_IN/ANT_OUT for the internal antenna (ST fig. 2).
5. Pin map checked against DS13258 table 15: NTC ADC1_IN6/8/9/10 (+ IN2 on PA7 for TH305), IRQs on WKUP1/WKUP4, I2C1, USART1. PB4 = 3V3_SLEEP: PB4 resets as NJTRST with pull-up, so the 3V3 rail boots in normal mode; firmware makes PB4 a GPIO output before using sleep mode. PB5 = LED, PA6 = GAUGE_BTP.
6. R307 removes the LED for sleep-current measurements.
7. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
8. SW301 resets the MCU. TP306 is BOOT0: hold it at 3V3 while releasing reset for the UART bootloader.
9. R313 (DNP) holds 3V3_SLEEP high (normal mode) while the MCU is off. TP307 probes it.
10. R312 feeds the SMBus pull-ups: remove it and supply TP308 for EV2400 work while 3V3 is down.
11. R316–R319: J303 can override NTC_EN/EN_5V1/EN_5V2/CHG_INH without shorting the MCU outputs.
12. PB8 = CHG_INH (rev A5): drive LOW to allow charging. High or hi-Z stops both chargers (R417/R517 pull CE up to 3V3), so reset/hang/unprogrammed stops charging. Low only while charging (69 uA per port), high in idle. No Standby.
13. Watchdog: IWDG in hardware mode (option byte IWDG_SW = 0) restarts a hung MCU; it runs from LSI, also in Stop/Standby.

STATUS LED





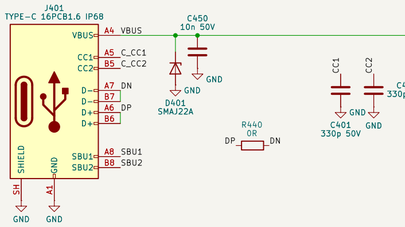
Not reviewed, laid out, built or tested
Spec + open items: DevBoard/README.md
STM32WB1MMC, debug headers, NTC TH1–TH4
Monolith 100 W battery bank — dev board

Sheet: /MCU and NTCs/
File: mcu.kicad_sch

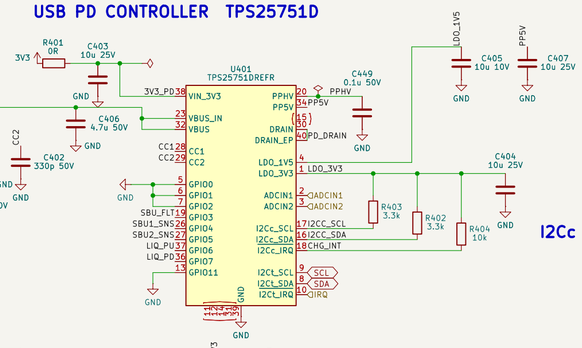
Title: MCU and board NTCs

Size: A3	Date: 2026-10-11	**Rev: A (draft)**
KiCad E.D.A. 10.0.7		Id: 4/6

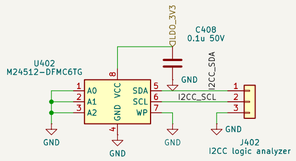
USB-C CONNECTOR + PROTECTION



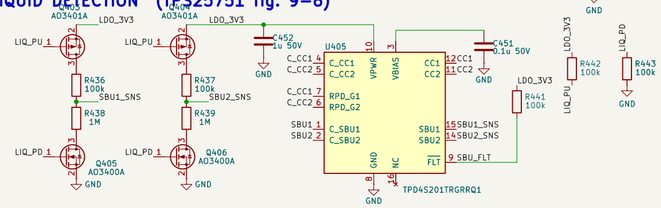
USB PD CONTROLLER TPS25751D



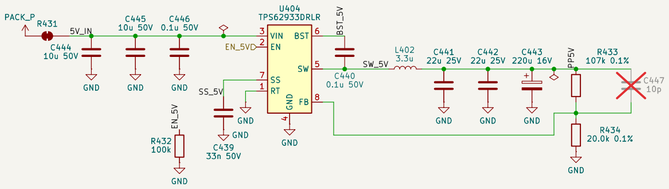
I2Cc BUS: EEPROM + PULL-UPS



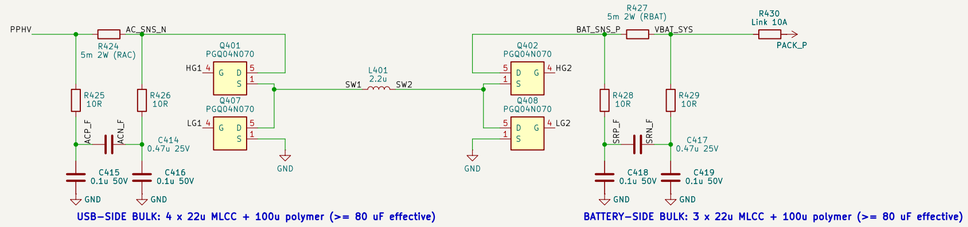
LIQUID DETECTION (TPS25751 fig. 9-8)



PP5V SUPPLY TPS62933 (1.2 MHz, 5.08 V) - 5 V source + VCONN

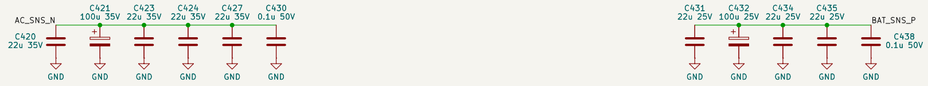


BQ25756 4-SWITCH BUCK-BOOST POWER STAGE (copy production layout exactly)

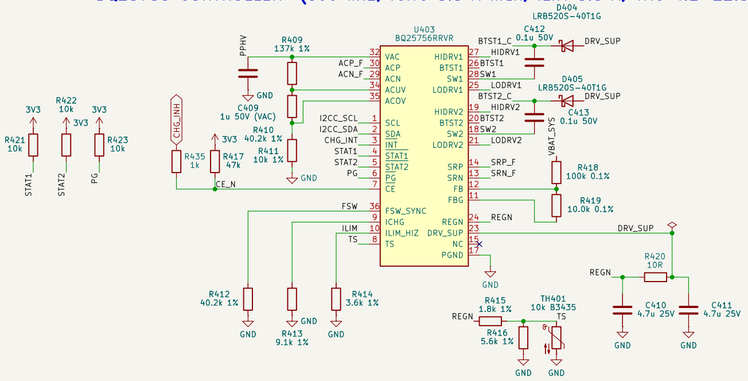


USB-SIDE BULK: 4 x 22u MLCC + 100u polymer (>= 80 uF effective)

BATTERY-SIDE BULK: 3 x 22u MLCC + 100u polymer (>= 80 uF effective)



BQ25756 CONTROLLER (600 kHz, ICHG 5.5 A max, ILIM 5.6 A, VAC 4.1-22.5 V window)



GATE / BOOTSTRAP TUNING (BQ25756EVM fig. 3-1: R8, R20, R3, R17, R13, R23, snubbers)



TEST POINTS / POWER FLAGS

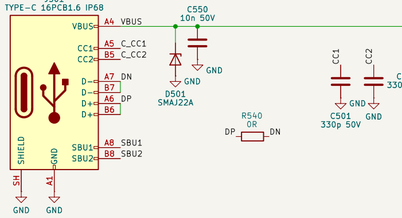


NOTES (this sheet is used twice: USB Port 1 = 4xx, USB Port 2 = 5xx)
1. Address straps are on the root sheet: port 1 ADCIN1 = LDO_3V3, ADCIN2 = GND (0x20);
   port 2 ADCIN1 = ADCIN2 = GND (0x21). Both are SafeMode (TPS25751 table 8-6).
2. Unused TPS25751 GPIOs (0, 1, 2, 11) tied to GND per datasheet (KiCad ERC flags this: expected).
   Never configure them as outputs.
3. CC and SBU are protected by U405 (TPD4S201, below): C401/C402 on the controller side. D+/D- are shorted by R440 (passive BC1.2 DCP); the TPS25751
   no longer does BC1.2. Liquid detection (TI fig. 9-8): GPIO4/5 read SBU1/2 through U405 (TPD4S201: CC and SBU OVP + ESD);
   GPIO6 = liquid_pmos_control (LIQ_PU), GPIO7 = liquid_nmos_control (LIQ_PD) in the TPS25751 configuration.
   U405 FLT -> GPIO3 = Fault_Input_Event (SBU_FLT, R443): an SBU short to VBUS forces error recovery.
   R442/R443 hold the liquid-detect FETs off while GPIO6/7 are Hi-Z.
4. Kelvin-route ACP/ACN from R424 and SRP/SRN from R427: filter parts at the IC.
5. Two MLCCs of each bulk bank plus C430/C438 at the FETs: the polymer (C421/C432) and remaining MLCCs
   one row back. Polymer carries the bulk (no DC-bias loss). MLCCs carry the switching-frequency ripple.
6. I2Cc pull-ups go to this TPS25751's own LDO_3V3 (TI guidance).
7. R430 = 10 A copper link to PACK_P (net-tie footprint, not a part); widen with zones.
8. VAC is taken before R424 per TI typical application; DRV_SUP fed from REGN via R420.
9. PPHV bulk (BQ25756 input bank) exceeds the 100 uF USB PD c5nkBulkPd: the TPS25751D sink switch soft-start limits
   inrush (C x SS: about 125 uF effective x 0.47 V/ms max = 0.06 A at SS setting 0).
   Set PPHV soft-start to setting 0 or 1 in the TPS25751 configuration.
10. Diamonds are PWR_FLAGs (ERC only, no part): each sits where its rail is sourced through a passive part.
11. CHG_INH (rev A5): R417 pulls CE up to 3V3, so charging needs the MCU to drive CHG_INH low through R435.
    Reset, IWDG reset, a hang or an unprogrammed MCU stop charging: a dead pack (3V3 down) still charges. Keep DIS_CE_PIN = 0.
12. Tuning block (review 3): R444-R449 0R in the gate and BTST lines, R450/C454 and R451/C455 DNP snubbers.
    Place the 0R right at the FET gates / BTST pins: they are where a few ohms go if the edges ring.

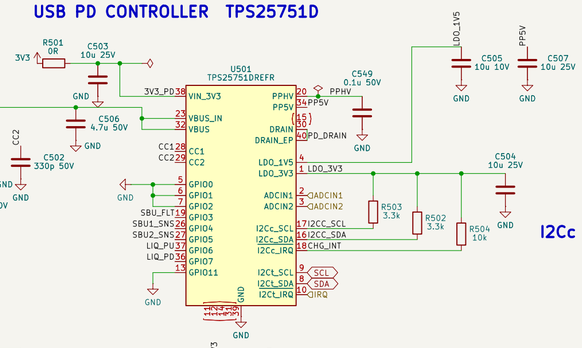
Not reviewed, laid out, built or tested
Spec + open items: DevBoard/README.md
TPS25751D + BQ25756 + TPS62933; port 1 = 4xx, port 2 = 5xx
Monolith 100 W battery bank — dev board

Sheet: /USB Port 1/		
File: usb_port.kicad_sch		
Title: USB-C PD port (used twice)		
Size: A2	Date: 2026-10-11	**Rev: A (draft)**
KiCad E.D.A. 10.0.7		Id: 5/6

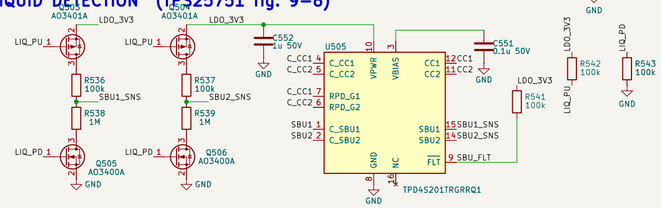
USB-C CONNECTOR + PROTECTION



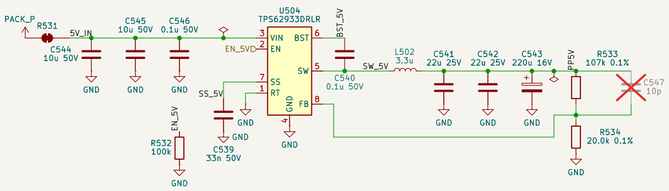
USB PD CONTROLLER TPS25751D



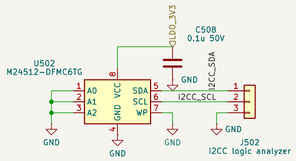
LIQUID DETECTION (TPS25751 fig. 9-8)



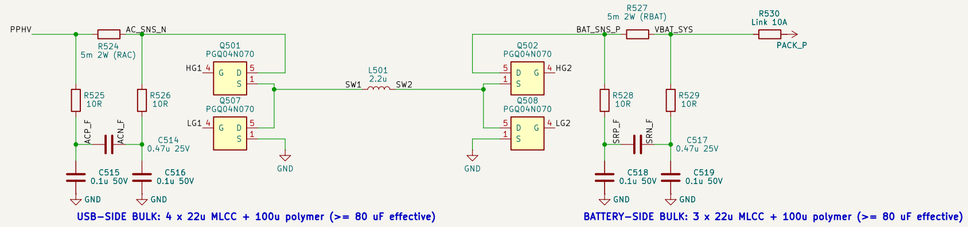
PP5V SUPPLY TPS62933 (1.2 MHz, 5.08 V) - 5 V source + VCONN



I2Cc BUS: EEPROM + PULL-UPS

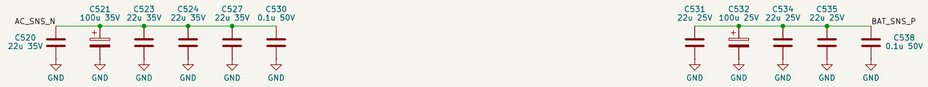


BQ25756 4-SWITCH BUCK-BOOST POWER STAGE (copy production layout exactly)

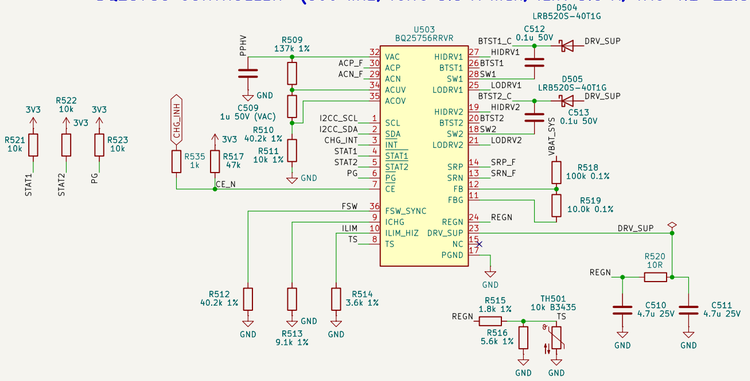


USB-SIDE BULK: 4 x 22u MLCC + 100u polymer (>= 80 uF effective)

BATTERY-SIDE BULK: 3 x 22u MLCC + 100u polymer (>= 80 uF effective)



BQ25756 CONTROLLER (600 kHz, ICHG 5.5 A max, ILIM 5.6 A, VAC 4.1-22.5 V window)



GATE / BOOTSTRAP TUNING (BQ25756EVM fig. 3-1: R8, R20, R3, R17, R13, R23, snubbers)



TEST POINTS / POWER FLAGS



NOTES (this sheet is used twice: USB Port 1 = 4xx, USB Port 2 = 5xx)
1. Address straps are on the root sheet: port 1 ADCIN1 = LDO_3V3, ADCIN2 = GND (0x20);
   port 2 ADCIN1 = ADCIN2 = GND (0x21). Both are SafeMode (TPS25751 table 8-6).
2. Unused TPS25751 GPIOs (0, 1, 2, 11) tied to GND per datasheet (KiCad ERC flags this: expected).
   Never configure them as outputs.
3. CC and SBU are protected by U405 (TPD4S201, below): C401/C402 on the controller side, D+/D- are shorted by R440 (passive BC1.2 DCP); the TPS25751
   no longer does BC1.2. Liquid detection (TI fig. 9-8): GPIO4/5 read SBU1/2 through U405 (TPD4S201: CC and SBU OVP + ESD);
   GPIO6 = liquid_pmos_control (LIQ_PU), GPIO7 = liquid_nmos_control (LIQ_PD) in the TPS25751 configuration.
   U405 FLT -> GPIO3 = Fault_Input_Event (SBU_FLT, R443): an SBU short to VBUS forces error recovery.
   R442/R443 hold the liquid-detect FETs off while GPIO6/7 are Hi-Z.
4. Kelvin-route ACP/ACN from R424 and SRP/SRN from R427: filter parts at the IC.
5. Two MLCCs of each bulk bank plus C430/C438 at the FETs: the polymer (C421/C432) and remaining MLCCs
   one row back. Polymer carries the bulk (no DC-bias loss). MLCCs carry the switching-frequency ripple.
6. I2Cc pull-ups go to this TPS25751's own LDO_3V3 (TI guidance).
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    Place the 0R right at the FET gates / BTST pins: they are where a few ohms go if the edges ring.

Not reviewed, laid out, built or tested
Spec + open items: DevBoard/README.md
TPS25751D + BQ25756 + TPS62933; port 1 = 4xx, port 2 = 5xx
Monolith 100 W battery bank — dev board

Sheet: /USB Port 2/
File: usb_port.kicad_sch

Title: USB-C PD port (used twice)

Size: A2	Date: 2026-10-11	**Rev: A (draft)**
KiCad E.D.A. 10.0.7		Id: 6/6